Shopping Cart

1. **Product Detection**
2. **retrieves product details from a database**
3. **Show cart contents (name, price, quantity, total)**
4. **Li-Po battery**
5. **buck converter for stable supply.**
6. **Voltage regulators to protect sensitive components.**

Checkout counter

1. **At the counter, the cart connects via Wi-Fi, Bluetooth, or NFC.**
2. **The computed cart details are sent to the store's billing system.**
3. **The counter system receives the synchronized data.**
4. **It generates a digital or printed receipt instantly**
5. **Customer pays via QR code, NFC card, or e-wallet.**
6. **Transaction is logged in the store's database.**
7. **Optional RFID/barcode gate ensures items match the receipt.**
8. **Prevents errors or theft.**

Smart Grocery Shelf

1. **Detect if items are removed or replaced**
2. **Ensure that both cart and shelf agree on product movement**
3. **Updates stock levels instantly**
4. **Aligns with automated checkout so receipts reflect accurate inventory**
5. **Restocking alerts and demands forecasting**
6. **Digital price tags for dynamic pricing, promotions, or nutritional info.**
7. **Ensures promotion or personalized recommendation**
8. **Smart shelves track item removal to prevent theft or misplacement**
9. **Cross checked with cart data before checkout**

Warehouse monitoring

1. **Update stock levels instantly**
2. **Syncs with smart shelves to have reflect information**
3. **Prevents "out-of-stock" surprises at checkout**
4. **When a cart confirms purchase it should be automatically deducted to warehouse listings**
5. **It should predict restocking and demand forecasting**
6. **It should generates alerts for low stock, expired items, or misplacements**
7. **Warehouse monitoring should feeds into the same database as carts and shelves**
8. **Provides dashboards for managers to oversee stock, sales, and customer trends**
9. **Aligns with exit validation at checkout to prevent theft or mismatch billing**

Parking lot system

1. **Cars entering are tag or scanned**
2. **Syncs with smart carts so the system knows which customer session to link purchases to,**
3. **Detect the available parking slots**
4. **Data should be linked to smart shelves and carts to predict customer traffic demand**
5. **Parking is merged with automated checkout receipts**
6. **Parking fees can be merged with automated checkout receipts**
7. **Customers pay for both groceries and parking in one transaction**
8. **Parking lot data feeds into the same database.**
9. **Exit gates validate that the carts receipt matches the items carried out.**
10. **Link the checkout confirmation with parking exit clearance.**

Smart supermarket dashboards

1. **Combines warehouse monitoring with smart shelf data.**
2. **Real-time stock levels, expiry alerts, and restocking**
3. **Integrates parking lot, shopping cart, and restocking**
4. **Predict peak hours and events**
5. **Display total sales, payment methods, and checkout speed.**
6. **Aligned with checkout receipts**
7. **Alert staff if a cart , shelf, gate malfunctions.**